1st Order DEs: Sketching Solutions

Questions 1–16

Question 1. Sketch the graph of $y = x^3 + c$ for $c = -1, c = 0$, and $c = 1$ on a single set of axes.

Question 2. Sketch the graph of $y = \frac{1}{x} + c$ for $c = -1, c = 0$, and $c = 1$ on a single set of axes.

$\frac{dy}{dx} = \mathrm{f}(x)$

Question 3. It is given that

$$\frac{dy}{dx} = 2x$$

a) For what value of x is the gradient zero?
b) Find the general solution to the differential equation.
c) Sketch the family of solutions represented by the general solution on a single set of axes.
d) Describe how the family of solutions are related geometrically.

Question 4. It is given that

$$\frac{dy}{dx} = \cos x$$

a) Find the general solution to the differential equation.
b) Sketch the family of solutions represented by the general solution on a single set of axes.

Separation of Variables

Question 5. It is given that

$$\frac{dy}{dx} = 3y$$

a) Show that $y = Ae^{3x}$, where A is a constant.
b) Sketch the family of solutions represented by the general solution when $A < 0$ and when $A > 0$ on a single set of axes.
c) Describe what happens as $x \to -\infty$

Question 6. It is given that

$$\frac{dy}{dx} = y - 2$$

a) Determine the constant solution to the differential equation.
b) Find the general solution to the differential equation.
c) Sketch the family of solutions represented by the general solution on a single set of axes.

Question 7. Solve each of these differential equations and hence sketch the family of solutions.

a) $\frac{dy}{dx} = -2y$

b) $\frac{dy}{dx} = 1 - y$

c) $\frac{dy}{dx} = y^2$

d) $\frac{dy}{dx} = \frac{1}{2y}, \quad y > 0$

Prerequisite Knowledge for Q8: A-Level Maths Partial Fractions

Question 8. It is given that

$$\frac{dy}{dx} = y(1 - y)$$

a) Determine the constant solutions to the differential equation.

b) Use separation of variables to show that

$$y = \frac{Ae^x}{1 + Ae^x}$$

c) Explain why one of the constant solutions from part a) does not appear in the solution found in part b).

d) Show that the solution in b) can also be written in the form

$$y = \frac{1}{1 + Be^{-x}}$$

e) Explain why the other constant solution from part a) does not appear in the solution found in part d).

f) Sketch the family of solutions represented by the solutions in parts a) and b) on a single set of axes.

Integrating Factors

Question 9. It is given that

$$\frac{dy}{dx} + y = 3$$

a) Find the constant solution of the differential equation.
b) Use an integrating factor to solve the differential equation.
c) Sketch the family of solutions on a single set of axes.

Question 10. It is given that

$$\frac{dy}{dx} - y = 2$$

a) Find the constant solution of the differential equation.
b) Solve the differential equation.
c) Describe the behaviour of solutions as $x \to \infty$
d) Sketch the family of solutions on a single set of axes.

Question 11. It is given that

$$\frac{dy}{dx} + 2y = 6$$

a) Find the constant solution of the differential equation.
b) Solve the differential equation.
c) Describe the behaviour of solutions as $x \to \infty$
d) Sketch the family of solutions on a single set of axes.

Prerequisite Knowledge for Q12-Q15: Reversing the Chain Rule, Integration by Parts

Question 12. It is given that

$$\frac{dy}{dx} + 2xy = 4x$$

a) Solve the differential equation.
b) Describe the behaviour of the solutions as $x \to \infty$ and as $x \to -\infty$
c) Sketch the family of solutions on a single set of axes.

Question 13. It is given that

$$\frac{dy}{dx} + y = x + 1$$

a) Use an integrating factor to show that the general solution is

$$y = x + Ae^{-x}$$

where A is a constant.

b) Show that the line $y = x$ is itself a solution of the differential equation.

c) Explain why every solution curve approaches the line $y = x$ as $x \to \infty$

d) Sketch the family of solutions on a single set of axes.

e) Show that distinct solution curves do not intersect.

Question 14. It is given that

$$\frac{dy}{dx} - y = x$$

a) Solve the differential equation.

b) Show that the line $y = -x - 1$ is a solution of the differential equation.

c) Describe the behaviour of the solutions as $x \to -\infty$

d) Describe the behaviour of the solutions as $x \to \infty$

e) Sketch the family of solutions on a single set of axes.

Question 15. It is given that

$$\frac{dy}{dx} + y = \sin x + \cos x$$

a) Solve the differential equation to show that $y = \sin x + Ae^{-x}$ for constant A
b) Explain why every solution approaches the curve $y = \sin x$ as $x \to \infty$
c) Sketch the family of solutions on a single set of axes.

Prerequisite Knowledge for Q16: Calculus with Hyperbolic Functions

Question 16. It is given that

$$\frac{dy}{dx} + y = 4\sinh x$$

a) Solve the differential equation to show that

$$y = e^x + (A - 2x)e^{-x}$$

where A is a constant.
b) State the coordinates of the point where a solution curve crosses the y-axis in terms of A
c) Describe the behaviour of the solutions as $x \to \infty$ and as $x \to -\infty$
d) Show that the stationary points of the family of solution lie on the curve

$$y = 4\sinh x$$

e) Sketch the family of solutions on a single set of axes.